MECH 151

Technical Design

Unit Reference Guide

A student quick-reference for every unit — Design Fundamentals through Design for Manufacturing — for portfolio work, competency checks, and the final prototype.

Foundations | Unit 1 | Unit 2 | Unit 3 | Final Exam

COURSE MAP

Semester Roadmap, Weeks 4–17

One project-based portfolio module per week; a short foundations unit precedes Unit 1.



- **Design Fundamentals**
- **Unit 1 — CAD Fundamentals**
- **Unit 2 — Design for Assembly**
- **Unit 3 — Design for Manufacturing**
- **Final Exam — Prototype Project**

How grading lines up: each unit maps to a graded portfolio module and closes with a unit competency check; the Design Portfolio (50% of the course grade) and the Prototype + Final Exam (30%) are graded separately — see the syllabus for the remaining weighting.

COURSE NOTE

This Course Is CAD-Program-Agnostic

The field of CAD software is too broad to teach every package — so this course teaches the operations that transfer between all of them.

Dozens of parametric CAD packages — are used across industry — SolidWorks, Fusion 360, Onshape, Inventor, Creo, NX, CATIA, and more

It's impractical to teach all of them in one course — and mastering only one would leave you stuck the moment a job, internship, or team uses something else

So this course teaches the operations every one of them shares — the specific menu names are yours to look up in whatever software you're using

The Operation	Sometimes Called
Push/pull a 2D profile into a solid	*Extrude · Pad · Boss / Cut*
Spin a profile around an axis	*Revolve · Rotate*
Lock a sketch so it can't drift	*Relations · Constraints*
The running history of modeling steps	*Feature Tree · Timeline · Model Tree*
Position parts relative to each other	*Mates · Joints · Assembly Constraints*
Round over or angle-cut a sharp edge	*Fillet / Chamfer · Round / Bevel*

Why it matters: everything in this deck — sketching, feature trees, mates, GD&T — describes the operation, not a specific menu path. Whatever software you're using in class, an internship, or a job, look for the tool that does the same job.

FOUNDATIONS

Design Fundamentals

Step back from the software: what a CAD model is for, before Unit 1's sketches and features begin.

BEFORE UNIT 1

FOUNDATIONS

The Design Process Is a Cycle

Every design loops — the first prototype is expected to be wrong, and that's the process working.



then back to Design (or Define), carrying what you learned

Define — state what the part must do, fit, withstand, and cost — before any geometry exists

Design — concept options, then the parametric models and drawings that pin down every dimension

Build — make a prototype — a print, a machined part, a mock-up

Test — measure it, fit it, load it, use it the way it will really be used

Evaluate — compare what happened against the requirements; loop back with what you learned

Why it matters: if you assume the first version is final, you build a model that's painful to change. If you assume you'll revise it, you build for iteration — clean sketches, captured intent, a readable feature tree.

FOUNDATIONS

Define the Problem Before You Design

The step skipped most often — and the one that wastes the most time when skipped.

S	M	A	R	T
Specific	**Measurable**	**Achievable**	**Relevant**	**Time-bound**
says exactly what, for whom	has a number you can check	possible with your time & tools	actually serves the need	due by a defined point

SMART in practice: "hold a phone at a comfortable viewing angle" is a direction. "Hold a 600 g phone at 55–70° without tipping, printable in under 4 hours, done by Week 12" is something you can design to and test against.

- Requirements, objectives, and constraints get lumped together as "the spec" — but they filter and rank designs differently (next: how each one works).

FOUNDATIONS

Requirements, Objectives, Constraints

Requirements and constraints filter designs to the acceptable ones; objectives rank the acceptable ones to find the best.

Requirements

What the design must do — pass/fail

- "Supports 5 kg without permanent deflection"
- Either met or not; missing one disqualifies a design
- Verified by test or inspection

Objectives

What you're optimizing — directions of "better"

- "As light as possible," "minimize cost"
- Not pass/fail — compared by degree
- Often trade against each other

Constraints

Fixed limits on the solution space

- "Fits a 150 mm cube," "$25 budget," "due Week 12"
- Not about function — about what's allowed
- Violating one disqualifies the design

Why it matters: mix them up and you either reject a good design for missing a "nice to have," or ship one that fails a real must-have.

Evaluating success — turn objectives into weighted, measurable criteria and score concepts with a decision matrix — screen pass/fail items first, then rank.

FOUNDATIONS

Knowing When to Stop: The 80/20 Rule

No design is ever perfect — chasing perfection is its own failure mode.

80/20

~80% of a design's useful performance comes from the first 20% of the effort — chasing the rest costs far more than it returns.

Move forward once requirements are met — even if you can see ways it could be nicer

Perpetual designing is a cost — every revision delays the prototype, the test, and everything you'd learn from them

"Good enough" is a decision — judge against written requirements, not an imagined ideal

Hold both ideas at once: expect the first version to be wrong, and resist the urge to perfect it. The cycle moves forward when a design is good enough to learn from — not when it's flawless, and not before it meets its requirements.

FOUNDATIONS

Design Intent

The reasoning a model should capture — what you expect to stay true when a dimension changes.

Sketch relations — symmetric, equal, concentric, collinear say why geometry is where it is, so it stays right when dimensions change

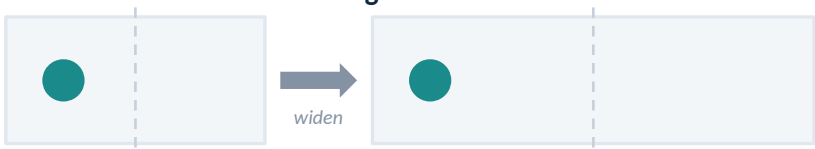
Which dimensions you choose — dimension from the reference that actually matters — a mounting face, a mating edge — not whatever's convenient

Feature order & references — build later features off stable geometry, not faces that may move or disappear in an edit

Equations & linked values — when two dimensions must always match or follow a rule, drive one from the other instead of typing both

Same result, different intent — dimension a hole two ways:

Dimensioned from the left edge:



→ widen the block and the hole drifts off-center

Tied to center by a symmetric relation:



→ widen the block and the hole stays centered

Why it matters: both models look identical until the block changes size — only the second one captures what you actually meant, not just a position that happened to look right.

1 **Editability**

a well-built model updates the way you intend

2 **Communication**

the next person can read the tree and see the logic

3 **Downstream reuse**

assemblies, drawings & manufacturing files inherit it

FOUNDATIONS

Organizing a Design

When a design has more than one part, where do the controlling dimensions live?

Bottom-Up

Design each part alone, then assemble

- Parts are independent — editing one doesn't ripple
- Simple to manage; easy to reuse parts
- Good for well-understood or off-the-shelf parts
- Weak when parts must share dimensions

Top-Down

Start from the assembly's shared geometry

- Shared dimensions live in one place and propagate
- Keeps mating parts fitting as the design evolves
- Good for new designs with moving interfaces
- More setup; unmanaged references get fragile

Tools for top-down — layout sketch (fixes key sizes before parts exist), skeleton/master model (shared reference geometry), in-context references (shape one part against another directly).

Rule of thumb: use the simplest structure that captures the intent — reach for top-down only where parts genuinely must share dimensions.

FOUNDATIONS

Practices That Keep a Design Robust

1 Fully define every sketch

an under-defined sketch can drift on an unrelated edit

2 Reference stable geometry

the origin and default planes never move; a filleted edge might vanish

3 Keep sketches simple

one idea per feature — complexity goes in the feature tree

4 Name & organize features

a tree of "Extrude7, Cut12" tells the next person nothing

5 Minimize dependency chains

fewer things a feature relies on means fewer ways it breaks

6 Model the design, not just the shape

if a rule governs the part, build the rule in

UNIT 1 · WEEKS 4–8

CAD Fundamentals

Sketching, views, CAD configuration, and the first parametric model — the foundation every later unit depends on.

COMPETENCY CHECK IN WEEK 8

UNIT 1

There Is No Single "Right" Model

The same shape can be built from a different base feature, sketch set, or feature order — with effectively infinite valid combinations.

What actually separates good from poor: two classmates can hand in completely different feature trees and both be correct. What separates a good model from a poor one is whether it captures design intent, rebuilds without errors, and stays easy to edit — not whether it matches a canonical answer.

Focus on — making deliberate choices you can explain, not on guessing the "expected" answer.

This unit builds — the drawing and modeling foundation every later unit depends on — a controlled sketch, standard views, and a parametric solid that updates when you change it.

UNIT 1

The Origin: Where Every Model Begins

Every coordinate, sketch, and feature ultimately measures from one fixed point in space.

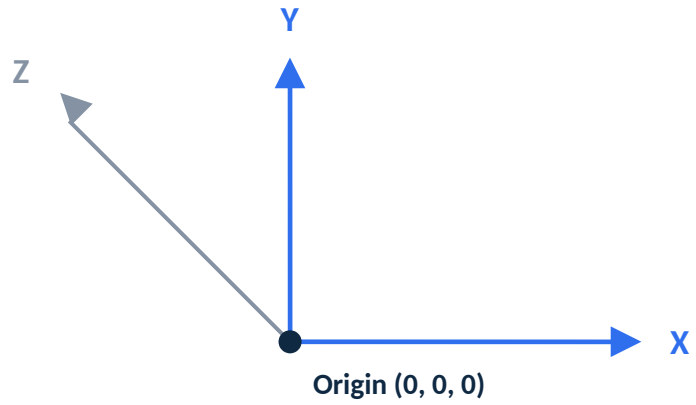
The origin (0, 0, 0) — the one point in the file that never moves, no matter how the model is edited

Three default planes — Front, Top, and Right all pass through the origin and split space along the X, Y, and Z axes

Everything is measured from it — directly or indirectly — a sketch plane, a mate, even a drawing's projected views all trace back to this fixed reference

Why it matters — reference the origin (or geometry built from it) instead of an edge that might move, get filleted away, or get deleted in a later edit

The three default planes meet here



Why it matters: the exact pairing of planes to axes (which plane is "Front") varies by CAD program — but every parametric system anchors its geometry to an origin and a set of default planes the same way.

UNIT 1

From Point to Solid: Building Up in Dimensions

Every model is a stack of dimensions — each one built from the last.

0D



Point

a location in space — no length, area, or volume. A sketch point, a vertex, the origin itself.

1D



Line / Curve

connects two points — has length only. Lines, arcs, and splines: what you actually sketch.

2D



Polygon (Profile)

a closed loop of 1D entities that encloses an area. A triangle, rectangle, circle — any closed profile.

3D



Solid (Volume)

push, spin, or sweep a profile through space and it gains volume. Extrude, revolve, sweep, loft.

Coming up: the next several slides walk through sketching (0D → 1D → 2D) and solid modeling (2D → 3D) in detail — this is the map for where each one fits.

UNIT 1

Creating a Body: The Feature Loop

Every solid feature repeats the same four steps — the feature tree is just this loop, run again and again.



repeat, starting the next loop from a face this feature just created

1 · **Pick a plane —** one of the three default planes, a flat face of the solid so far, or an offset/angled reference plane

2 · **Define a profile —** sketch a closed, fully-defined 2D profile on that plane — points and lines built into a polygon

3 · **Create a volume —** extrude, revolve, sweep, or loft the profile into a solid, then add, cut, or intersect it with what's already there

4 · **Dress it up —** apply fillets, chamfers, shells, drafts, or patterns — operations on the solid, not a new sketched volume

Why it matters: step 1 of the next loop is usually a face the previous loop just created, not a default plane — that dependency is exactly what builds the feature tree, one branch at a time.

UNIT 1

Sketching Fundamentals

A sketch is a 2D profile on a plane or face — the input to a 3D feature like an extrude or revolve.

1 Sketch entities

lines, arcs, circles, rectangles, slots, splines, and points

2 Reference geometry

the origin, the three default planes (front, top, right), and construction lines that guide without becoming part of the profile

3 Closed vs. open profiles

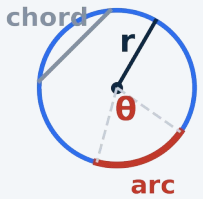
a closed loop can be extruded into a solid; an open profile makes thin features, surfaces, or paths

Why it matters: keep sketches simple — draw only the geometry the feature needs, keep profiles closed, and put complexity into separate features rather than one crowded sketch.

UNIT 1

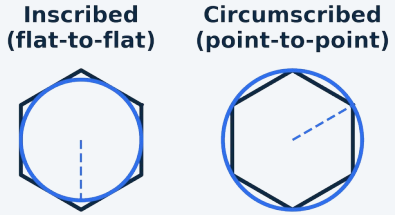
Geometry of Shapes: 2D Construction

Before a shape is a sketch, it's plane geometry — points, lines, and curves in defined relationships.



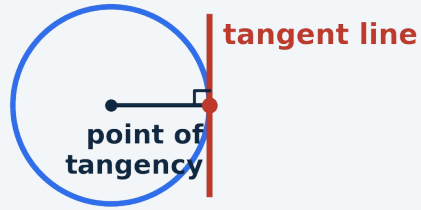
Circle & Arc Geometry

center + radius/diameter; a chord connects two points, a central angle subtends an arc.



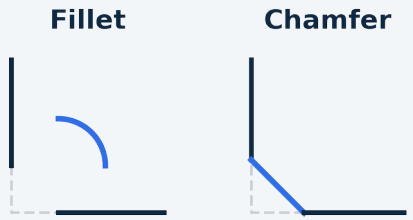
Polygons

inscribed (flat-to-flat) vs. circumscribed (point-to-point) — matters for hex features and bolt heads.



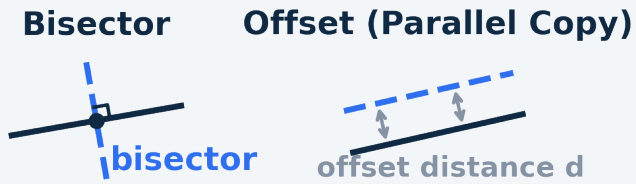
Tangency

a line/arc touches a circle at exactly one point, perpendicular to the radius there.



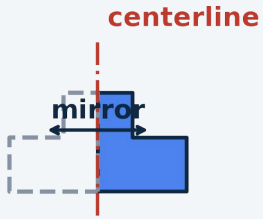
Fillets & Chamfers

a fillet is a tangent arc replacing a sharp corner; a chamfer is a straight cut across it.



Constructions

perpendicular bisector and offset — classic constructions, now done by relations.



Symmetry & Centerlines

build half a shape about a construction line and mirror it to guarantee identical halves.

UNIT 1

Sketch Constraints & Relations

Constraints lock down where sketch geometry can move, removing degrees of freedom until the sketch is fully defined.

Under-defined

- Geometry can still be dragged
- The sketch is not fully controlled

Fully defined

- Every entity fixed by relations + dimensions
- The target state before finishing a sketch

Over-defined

- Conflicting or redundant constraints
- Must be resolved, not ignored

Geometric relations — coincident, collinear, parallel, perpendicular, tangent, concentric, equal, horizontal, vertical, symmetric

Dimensions — linear, radial, diameter, angular

Why it matters: add relations before dimensions — relations capture intent (this hole stays centered) so changing one dimension updates the model predictably instead of distorting it.

UNIT 1

Orthographic Projection & Multiview Theory

Imagine the part inside a glass box — each face of the box catches one view; unfolding it gives the standard view arrangement.

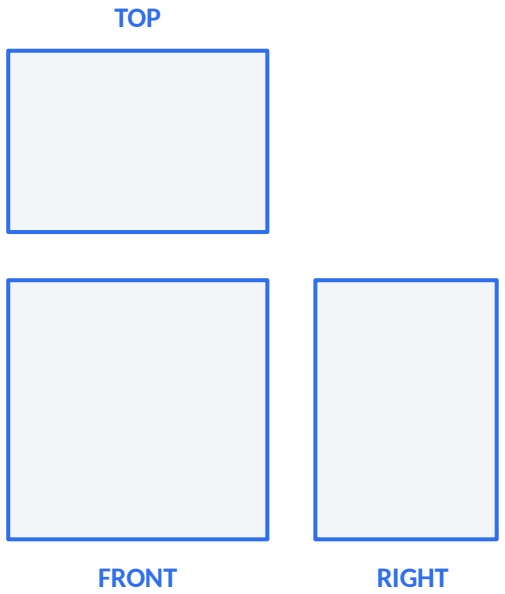
Third-angle projection — the U.S. standard — top view sits above front, right-side view sits to its right

Line conventions — visible edges solid, hidden edges dashed, centerlines for holes and symmetry axes

View selection — choose the front view with the most shape and fewest hidden lines; add only the views needed

Descriptive geometry — true length appears only when a line is parallel to the viewing plane; true shape only in a view perpendicular to that plane

Primary vs. secondary views — a primary view projects from a principal view; a secondary view projects from a primary view to reveal a feature still not true-size



UNIT 1

Basic Dimensioning Standards

Dimensions tell the reader how big the part is and how precisely each feature must be made (ASME Y14.5 in the U.S.).

1 Anatomy

extension lines, dimension lines, arrowheads, and the value — kept off the outline and out of the view

2 Placement

dimension to visible outlines, not hidden lines; group related dimensions; smallest closest to the view

3 Don't over-dimension

each feature is located once — a redundant dimension creates conflict on the shop floor

4 Feature callouts

diameter, radius, counterbore, countersink, and depth symbols for holes

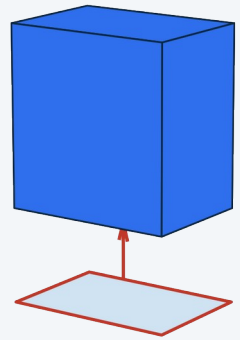
UNIT 1

Geometry of Shapes: Building 3D Solids

Every solid model is a running total: start with a volume, add more, and cut some away.

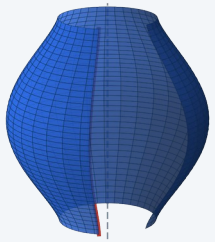
Extrude

push a profile along an axis



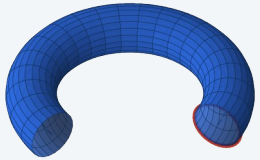
Revolve

spin a profile around an axis



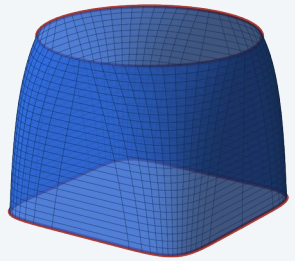
Sweep

run a profile along a path



Loft

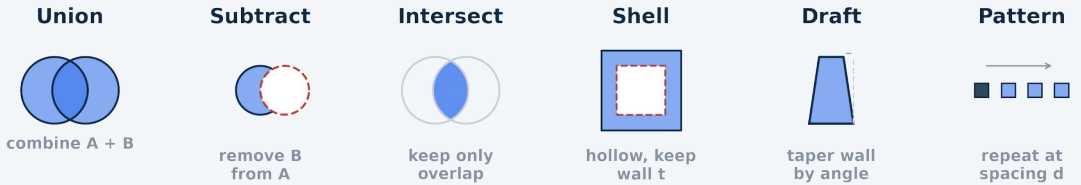
blend between different profiles



Primitives



Booleans & Dress-Up



Fillet & chamfer are shown on the sketching slide (2D construction).

UNIT 1

Introduction to Parametric Part Modeling

A parametric model is a history of features, each driven by dimensions and relations — change a dimension and the model rebuilds.

Base feature — the first solid, usually an extrude or revolve of a fully defined sketch

Added features — extrude, cut, revolve, fillet, chamfer, hole, shell, pattern — each added in order on the feature tree

Sketch planes & references — choosing where each feature's sketch lives so edits stay stable

Practice — editing driving dimensions, reordering/suppressing features, fixing rebuild errors — reading a feature tree and predicting what an edit will do

Why it matters: this is what makes CAD editable rather than something you redraw from scratch.

UNIT 1

Geometry of Shapes: Measurable Properties

Every shape carries numbers that follow from its geometry — the values a design is checked against.

Of a 2D profile

- Area, perimeter, centroid
- Second moment of area (I) — resistance to bending
- Section modulus (I ÷ distance to extreme fiber)

Of a 3D solid

- Volume, surface area
- Mass (volume × density), center of gravity
- Mass moment of inertia — resistance to angular acceleration

Garbage in, garbage out: the numbers are only right if the material density is set correctly and the model is a single watertight solid — an open or multi-body model gives a meaningless mass.

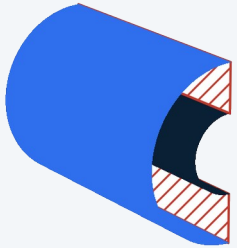
UNIT 1

Section Views

An imaginary cutting plane slices the part to show what's inside, replacing hidden-line tangles with a clear cut face.

Full Section

The cutting plane passes entirely through the part, exposing the interior along its whole length.



Half Section

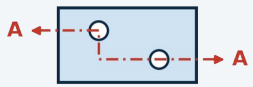
Only one quarter is removed — the exterior and the interior appear together in a single view. Best for symmetric parts.



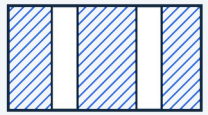
Offset Section

The plane is stepped (jogged) so it passes through features that aren't in line — each still appears true size.

Top view — stepped cutting plane



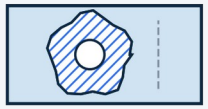
Section A-A — both holes shown true size



Local Sections

Broken-out, revolved, and removed — used to show one local feature without a full cut.

Broken-out



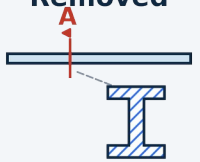
torn boundary reveals only the local interior

Revolved



true cross-section rotated 90° into the view, in place

Removed



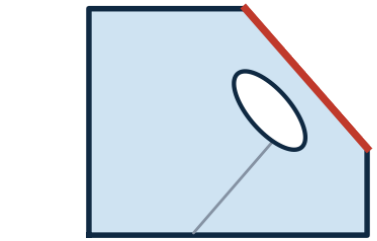
Convention: section lining marks the cut face — but ribs, webs, fasteners, shafts, and pins are never hatched, even when the cutting plane passes through them.

UNIT 1

Auxiliary Views

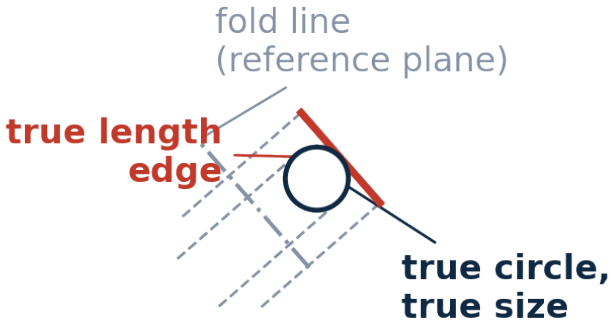
An inclined face looks foreshortened in every principal view — an auxiliary view is projected perpendicular to it so it appears true-size.

Front view — inclined face foreshortened



hole looks like an ellipse here

Primary auxiliary — projected true size

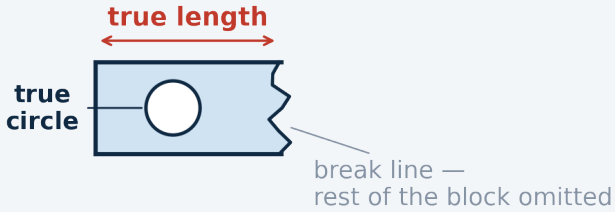


Primary auxiliary — projected from one principal view to show an inclined surface true-shape

Partial auxiliary — only the inclined feature is drawn — the rest is omitted to avoid distortion and clutter

Reference-plane method — measurements are transferred across the fold line along projection lines

Partial auxiliary — only the true-shape feature is drawn



UNIT 1

CAD Software Configuration

Consistent output comes from configured software, not from fixing every file by hand.

1 Templates

part, assembly, drawing templates with the right units, standard, title block

2 Document properties

units, dimensioning standard (ASME/ISO), grid, layers

3 Custom properties

file metadata that auto-populates the title block and BOM

4 File management

naming, folder structure, keeping references intact

UNIT 1

CAD/CAM Data Translation

Moving a model between systems means exporting to a neutral format — each keeps some information and loses the rest.

Solid exchange

STEP (AP203/214/242), IGES

- Geometry preserved
- Parametric feature history is not

Kernel formats

Parasolid, ACIS

- Used between packages sharing a modeling kernel

Mesh formats

STL, 3MF

- Faceted surface, no true curves
- Resolution set at export

Why it matters: lost or corrupted faces, gaps requiring geometry healing, and unit-scale errors are the common translation issues.

UNIT 1

Unit 1 Wrap-Up

1 Controlled sketches

fully defined, with relations that capture intent

2 Multiview drawing

correct projection, line types, and dimensions

3 First parametric model

a clean, editable feature tree

You'll produce: controlled sketches, a multiview drawing (with section and auxiliary views as needed), and a first fully parametric part model with correctly configured templates.
Assessed by: Unit 1 Competency Check (Week 8) and the corresponding portfolio modules.

UNIT 2 · WEEKS 8–11

Design for Assembly

Represent threads and fasteners, combine part models into working assemblies, document them, and design them to be quick and cheap to put together.

COMPETENCY CHECK IN WEEK 11

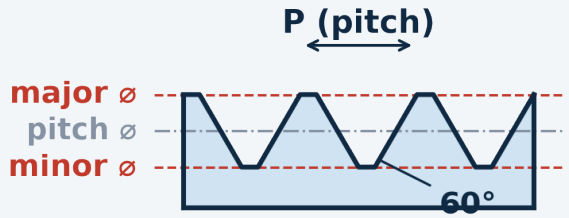
UNIT 2

Threads & Fastener Representation

Threads are almost never drawn in full detail — a standardized representation plus a note carries the information a shop needs.

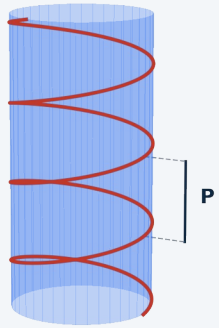
Thread Profile

Thread profile (external V-thread)



How a Thread Is Cut

One helical path, advancing one pitch per turn



Thread terms — major/minor diameter, pitch, lead, external vs. internal, right- vs. left-hand, class of fit

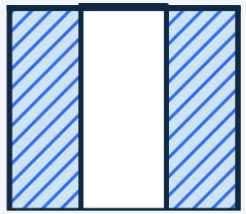
Representation — detailed, schematic, simplified — simplified is most common in practice

Fasteners — bolts, screws, nuts, washers, inserts — clearance and tap-drill holes are sized to fit them

Thread notes — 1/4-20 UNC-2A (inch) or M6 × 1 (metric) — diameter, pitch, and class in one callout

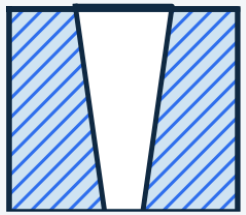
Common Hole Types

Through Hole



passes all the way through

Tapered Hole



diameter changes along its length

Blind Hole



stops before the far side (drill point)

Counterbore



flat-bottomed step for a bolt head

Countersink



conical seat for a flat-head screw

Spotface

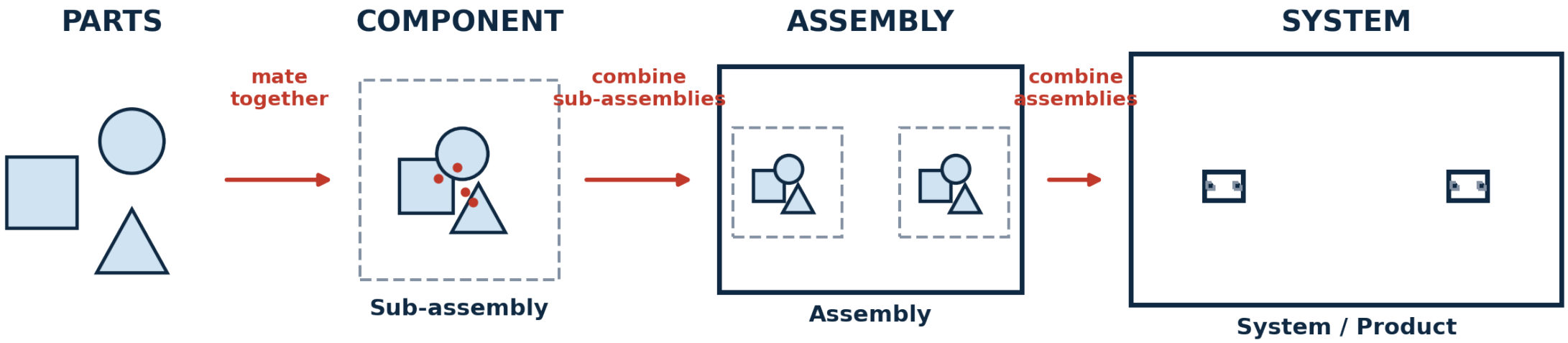


shallow flat land for a bearing surface

UNIT 2

From Parts to Systems

The same relationship repeats at every scale: individual parts mate into a component, components combine into an assembly, and assemblies combine into a complete system.



Why break it down: this is how large, complicated systems stay manageable — a vehicle or an instrument is never modeled as one flat list of thousands of parts. It's a tree of assemblies, each one a self-contained, testable unit.

UNIT 2

Assembly Modeling: Mates & Constraints

An assembly positions components relative to one another using mates — each removes degrees of freedom.

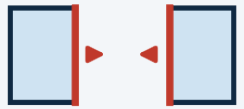
6

degrees of freedom a free rigid body has — mates remove them one axis at a time.

First component — the first part placed is fixed and becomes the ground the rest is built against

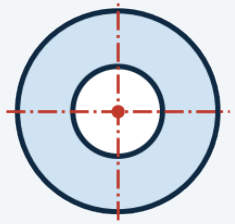
Motion — leaving rotational or linear freedom on purpose so a mechanism can be moved and checked

Coincident



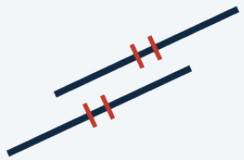
faces (or edges) forced flush

Concentric



shared axis — pin in a hole

Parallel



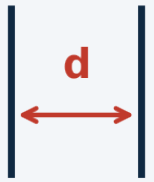
edges stay equidistant

Perpendicular



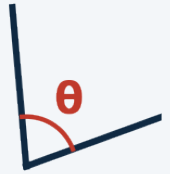
edges held at 90°

Distance



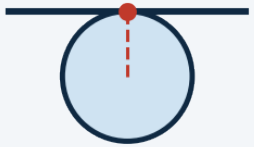
fixed gap between faces

Angle



fixed angle between edges

Tangent



curve just touches a line

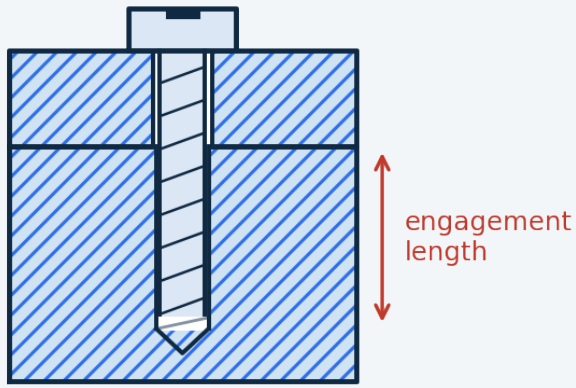
UNIT 2

Methods of Joining, Part 1

How two parts are held together is a design decision — choose deliberately, then design both parts to suit it.

Threaded Fasteners

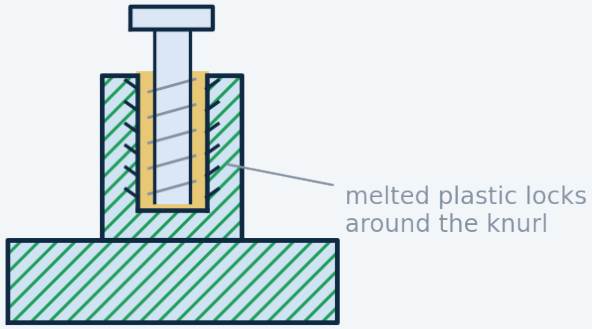
Into tapped holes



Default for metal-to-metal joints — strong, standard, and removable. Needs ~1.5–2× screw diameter of thread engagement plus tool access.

Heat-Set Inserts

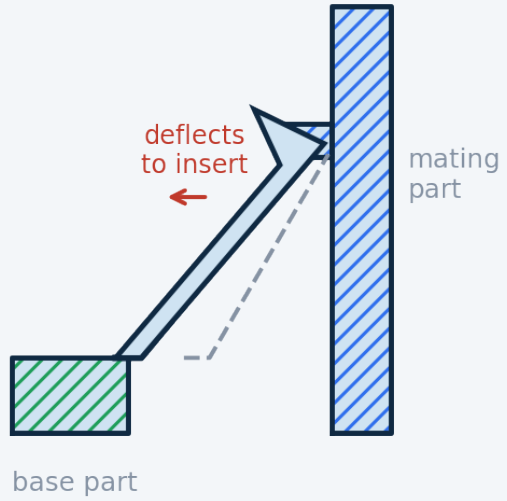
For plastic / 3D-printed parts



A brass insert is pressed into a boss with a soldering iron — melted plastic locks into its knurls, far more reusable than threading into plastic directly.

Snap-Fit Joints

No hardware, no tools



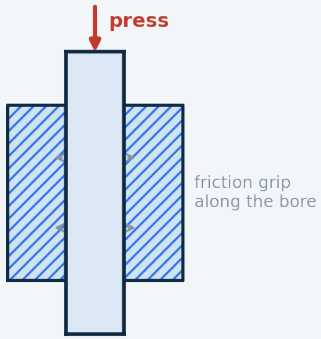
Cantilever, annular, and torsion snaps release or lock permanently depending on retention angle — design for allowable strain and a lead-in angle.

UNIT 2

Methods of Joining, Part 2

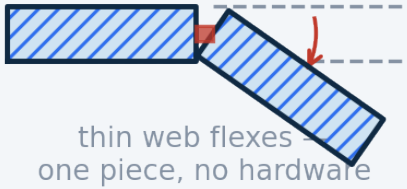
Press / Interference Fits

Hold by friction from a slight size mismatch between mating diameters — dowel pins, bushings, bearings. Cheap and precise, but hard to disassemble and tolerance-sensitive.



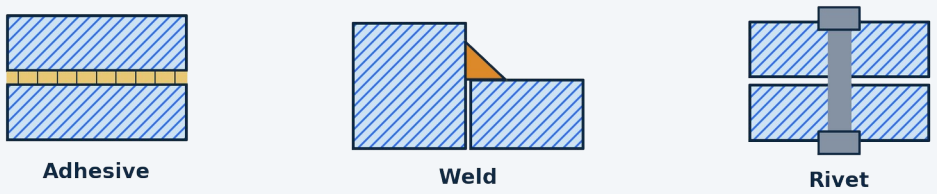
Compliant Mechanisms

Motion or retention comes from the material's own flexibility — living hinges, flexure pivots, one-piece clips. Cuts part count to the extreme; limited travel and fatigue life.



Permanent Joints

Adhesives, solvent/ultrasonic welding, and riveting — used only where the joint never needs to separate.



The DFA trade-off: hardware is universal and serviceable but adds parts, tools, and assembly time; snap-fits and compliant features add almost nothing to the parts list but must be designed carefully and are harder to revise once tooling is dialed in.

UNIT 2

Using Online Component Repositories

You should not model a part you can download — this is normal professional practice, not a shortcut.

Where models come from — McMaster-Carr (download STEP/native from almost any product page), manufacturer sites, TraceParts, GrabCAD, 3D ContentCentral

Why it matters for DFA — a downloaded model is the real catalog part — correct dimensions and thread, a real part number for the BOM, at the price and lead time listed

Bring them in cleanly — keep in a dedicated library folder, don't edit their geometry, mate to them like any other component

Check before you trust — confirm units/scale on import, simplify heavy models, verify the part number matches what you intend to buy

UNIT 2

Design for Assembly (DFA) Principles

Fewer parts, easier insertion, less to get wrong on the line — DFA turns assembly time into a design variable.

1 Minimize part count

a part is a candidate for elimination unless it moves, needs a different material, or must be removable

2 Prefer off-the-shelf parts

cheaper, faster, and more reliable than custom-fabricated — see the online repositories slide

3 Easy insertion & no reorientation

chamfers and self-locating features, assembling from one direction

4 Standardize fasteners & use symmetry

fewer unique sizes to stock and pick; make orientation obvious or irrelevant

Design review & iteration: the unit ends with a design review workshop — present an assembly, take feedback against the DFA principles, and produce a documented revision with a measurable improvement.

UNIT 2

Unit 2 Wrap-Up

1 Thread & fastener notes

correct callouts, class of fit, and hole sizing

2 Multi-part assembly

correct mates, sub-assemblies, and library hardware

3 Documented revision

a design review turned into a measurable DFA improvement

You'll produce: correctly called-out thread and fastener notes, a multi-part assembly model with correct mates, plus a documented design revision.
Assessed by: Unit 2 Competency Check (Week 11) and the corresponding portfolio modules.

UNIT 3 · WEEKS 12–16

Design for Manufacturing

State tolerances and fits, then close the loop from CAD model to physical part — manufacturing-ready files, process limits, and shop safety.

COMPETENCY CHECK IN WEEK 16

UNIT 3

Why Standards Carry the Design Intent

Once a design leaves your screen, the maker builds exactly what the documentation says.

The maker is not the designer — standard conventions (ASME Y14.5, GD&T, thread/finish callouts) let them read your intent without a phone call

Critical dimensions must be stated — a tolerance callout tells the shop where to spend effort and where a loose result is fine

Standard formats preserve meaning — STEP carries clean geometry between systems; a sloppy export arrives with gaps, wrong units, or lost features

Ambiguity is expensive and slow — on a graded prototype, it comes back as a part you can't use and no time to redo

Why it matters: capture the design intent, then communicate it in the form everyone downstream already knows how to read.

UNIT 3

Introduction to Tolerancing

No part is made to an exact dimension — tolerancing states how much variation is acceptable so parts still fit and function.

⌀	R	±	SQ	SR	°	25	( )
Diameter	**Radius**	**Plus/Minus**	**Square**	**Spherical Radius**	**Degrees**	**Basic Dimension**	**Reference Dim.**
⌀25	*R5*	*±0.01*	*SQ20*	*SR10*	*45°*	*[25]*	*(25)*

Limit Dimensions

1.253

1.247

high limit over low limit — no math on the shop floor

Plus/Minus Tolerance



1.250 ±0.003

bilateral shown; unilateral: +0.005 / −0.000

GD&T Callout



position, ⌀0.10 at MMC, to datum A

Close Fit



⌀25 H7/p6

near-zero gap — press or light-tap assembly (interference)

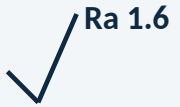
Loose Fit



⌀25 H11/c11

visible gap — free-running clearance, easy to assemble

Surface Finish



roughness average (µin/µm) — smaller number = smoother surface

UNIT 3

Tolerance Stackup: Bolt, Washer, Plate & Nut

Every part varies — a stackup adds those variations along a chain of parts to predict whether the assembly still works.

1 Define the gap
Pick the one dimension that decides fit or function — here, bolt protrusion P past the nut.

2 Draw the loop
Trace from one side of the gap to the other through every part in the chain.

3 Assign directions
A dimension that makes the gap bigger is (+); one that makes it smaller is (−).

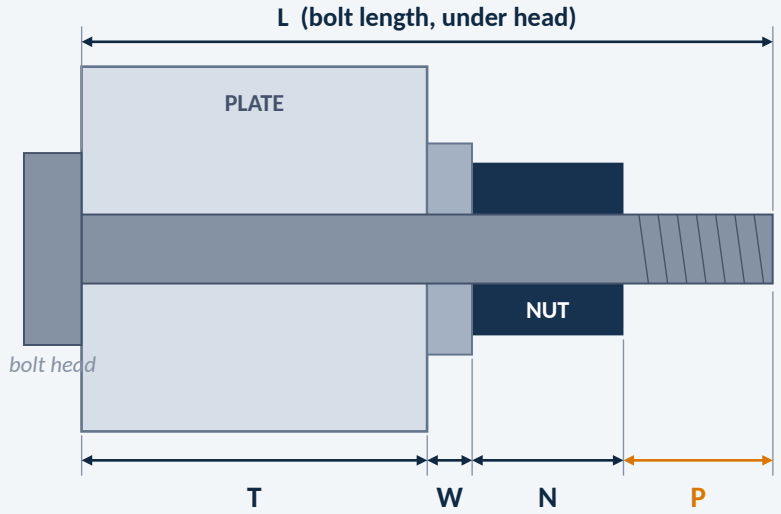
4 Sum the nominals
P(nom) = Σ(+) − Σ(−).

5 Sum the tolerances
Worst case: add every ± directly. Statistical (RSS): √(Σ tol²).

6 Compare to the requirement
Outside the limits? Tighten a tolerance or change a nominal — then re-run.

Section view

P = L − T − W − N



Part	Dim	Nominal (in)	Tolerance	Effect on P
Bolt, ¼-20 hex	L	1.000	±0.020	+ (longer → more P)
Plate	T	0.500	±0.010	−
Flat washer	W	0.065	±0.010	−
Hex nut	N	0.219	±0.007	−

UNIT 3

Tolerance Stackup: Worked Example

Requirement: 0.100 ≤ P ≤ 0.250 in. — at least 2 threads (2 × 0.050 pitch) past the nut, without hitting the cover above it.

Dim	Dir	Nominal	± Tol	Tol²
L bolt	+	+1.000	0.020	0.000400
T plate	−	−0.500	0.010	0.000100
W washer	−	−0.065	0.010	0.000100
N nut	−	−0.219	0.007	0.000049
Total		**P = 0.216**	**Σ = 0.047**	**Σ = 0.000649**

RSS tolerance = √0.000649 = ±0.025

Worst Case

× FAILS max by 0.013

P = 0.216 ± 0.047

Range 0.169 – 0.263 in. *Guarantees 100% of parts fit.*

Statistical (RSS)

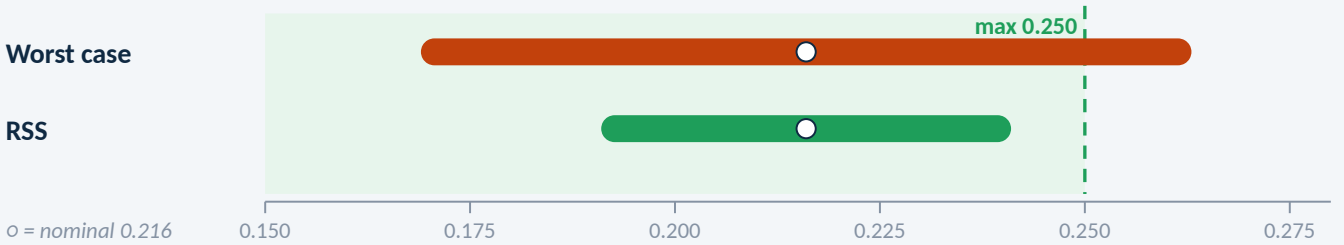
✓ PASSES

P = 0.216 ± 0.025

Range 0.191 – 0.241 in. *~99.7% fit if each ± is a 3σ process.*

Predicted protrusion vs. requirement (in.)

green = acceptable; min 0.100 is off-scale left



Fixing the worst case: tighten the biggest contributors first — bolt to ±0.010 and washer to ±0.005 gives Σ = 0.032, so P = 0.184 – 0.248 ✓.

Which method? Worst case for low volume or safety-critical fits; RSS for production runs where every part landing at its limit at once is unlikely.

UNIT 3

GD&T: The Building Blocks

A symbolic language (ASME Y14.5 / ISO GPS) for functional relationships — design intent written in symbols instead of prose.

Feature of size — a feature with an axis or centerplane — a hole, pin, slot, width

Datums (A, B, C) — reference surfaces chosen to match how the part actually seats in use

Datum reference frame — primary + secondary + tertiary datums lock all six degrees of freedom

Tolerance zone — the geometric space the feature must lie within — a gap, a cylinder, a band

Symbol families — the standard's ~14 geometric characteristic symbols sort into five families — form, orientation, location, profile, and runout (next slide)

Feature control frame — read left to right



Symbol (position) | Tolerance value | Datums referenced

Why it beats plain ±: a ± box on four hole dimensions creates a small square zone and over-constrains the part; a position callout defines a round zone tied to the surfaces that matter, letting more good parts through.

UNIT 3

GD&T Symbol Categories

The standard's ~14 geometric characteristic symbols sort into five families by what they control and whether they need a datum.

Form

No datum needed

- Straightness
- Flatness
- Circularity
- Cylindricity

Orientation

Needs a datum

- Perpendicularity
- Parallelism
- Angularity

Location

Needs datums

- Position
- Concentricity
- Symmetry

Profile & Runout

Profile: no/some datums · Runout: datum axis

- Profile of a line
- Profile of a surface
- Circular runout
- Total runout

Material condition modifiers: MMC (maximum material condition) and LMC (least material condition) let a position tolerance earn bonus tolerance as the hole moves away from its worst-case size — matching how a bolted clearance-hole joint really behaves.

UNIT 3

Manufacturing-Ready Files

Each process consumes a different file — a good part starts with exporting the right one at the right settings.

Additive

- Watertight mesh: STL or 3MF
- Fine enough to hide facets, not so fine the file is unwieldy

Laser Cutting

- 2D vector profile (DXF)
- Cut and engrave lines on separate layers

CNC Machining

- Solid model (STEP) + dimensioned drawing
- Drawing tells the machinist which dimensions are critical

CAM basics — stock setup, toolpath generation, simulation, and the post-processor that turns toolpaths into machine G-code.

UNIT 3

Machine Safety, PPE & Operation

Students are checked out on safe setup and operation before running any shop equipment.

Baseline PPE — ANSI Z87.1 safety glasses, closed-toe shoes, no loose clothing/jewelry/unrestrained hair

Rotating machinery — never wear gloves near a spindle, chuck, or bit — a caught glove pulls the hand in

Laser cutter — never run unattended; keep a view of the cut; know the extraction and fire response

3D printers — hot nozzle and bed — let them cool, don't reach into an active machine

Before starting — know the location of the e-stop, fire extinguisher, and first-aid kit

PPE is not optional: per the syllabus, a participant without proper PPE during machine use will be asked to leave the class for that session.

UNIT 3

Additive Manufacturing (3D Printing)

The course uses material-extrusion printing (FDM/FFF), building one deposited layer at a time.

Materials — PLA, PETG, ABS — printability, strength, and temperature trade-offs

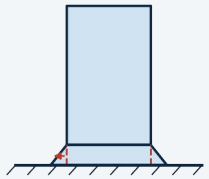
Orientation — layers are weakest in the build (Z) direction — a strength decision, not just fit-on-bed

Supports & overhangs — the roughly 45° rule, bridging limits, designing to avoid support

Settings that matter — layer height, wall count, infill density and pattern

Tolerances — holes print undersize, first layers spread — allow for it or drill/ream after

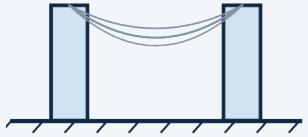
Common Print Issues & Fixes



Elephant Footing

the first layer over-extrudes and squishes outward under the nozzle

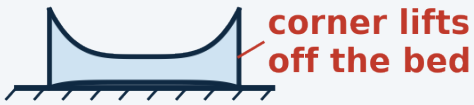
Fix — add a brim, or reduce first-layer horizontal expansion



Stringing

molten filament oozes from the nozzle while traveling between features

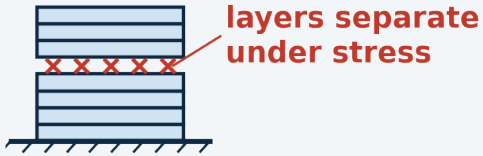
Fix — dry the filament; tighten retraction distance/speed



Warping

outer edges cool and shrink faster than the center, curling corners up

Fix — add a brim/raft; enclose the printer or use a heated bed



Layer Separation

layers don't fully fuse — often too cold, or cooled by a draft

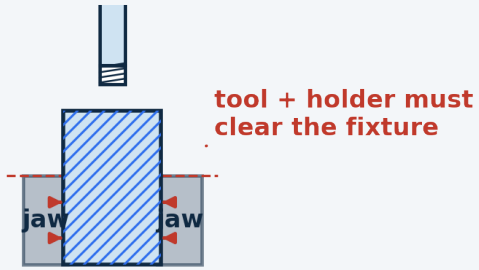
Fix — raise nozzle temp slightly; shield the print from drafts

UNIT 3

Subtractive Manufacturing (CNC)

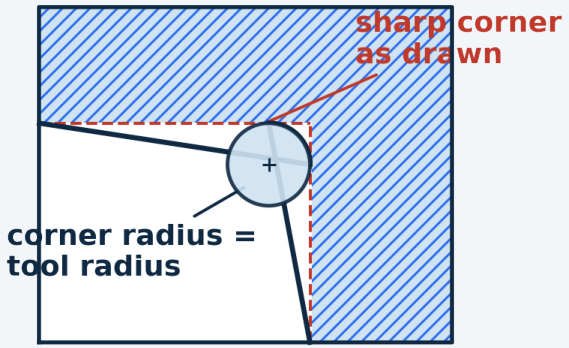
Starts with solid stock and cuts material away — design constraints come from the tool and how the part is held.

Workholding: rigid, and clear of the tool

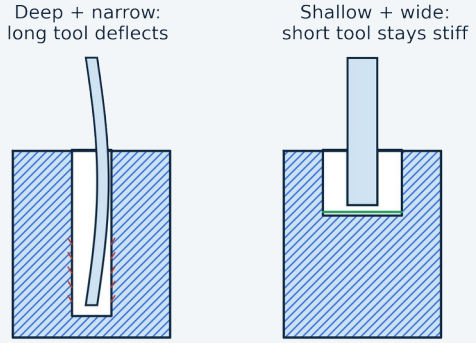


workpiece clamped between vise jaws

Internal corners inherit the tool radius



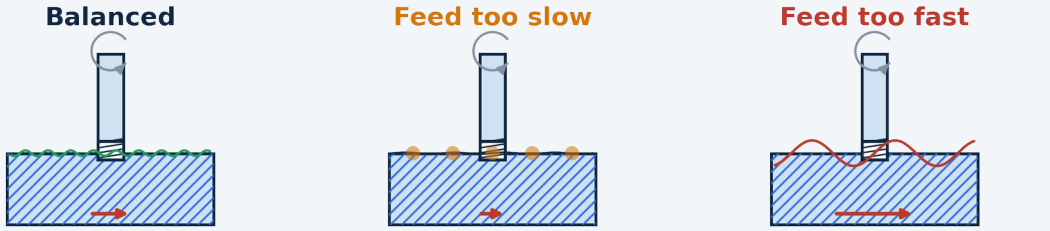
Tool reach vs. stiffness



long, thin tool flexes under load — chatter & tapered walls

short, stiff tool — straight walls, clean finish

Feeds & Speeds



even chip load — clean, consistent finish

tool rubs instead of cutting — heat builds, edges burn/wear fast

chip load too heavy — tool deflects, leaves chatter marks

UNIT 3

An Introduction to G-code

A plain-text list of moves the machine runs one line at a time — the shared language of CNC mills, 3D-printer slicers, and laser cutters.

- **What it is —** a text file a controller runs top to bottom; G words set motion and modes, M words switch machine functions
- **A line is a block —** a list of words, each an address letter plus a value: X25.0 is address X, value 25.0
- **Modal settings stick —** units, positioning, feed and speed stay in effect until a later line changes them
- **Where it comes from —** CAM or a slicer writes it; a post-processor tailors it to your exact machine — then you set the part zero and dry-run before cutting

One block, broken down



Modal words hold: this F150 and G01 keep acting on every line below until one is changed.

Codes you'll actually see

G20 / G21	inch / millimetre	**G02 / G03**	clockwise / CCW arc	**M03 / M05**	spindle on CW / off
G90 / G91	absolute / incremental	**G54–G59**	work offset (part zero)	**M06**	tool change
G17	work in the XY plane	**G28**	return to machine home	**M08 / M09**	coolant on / off
G00	rapid — non-cutting move	**G04**	dwell — timed pause	**M30**	end of program
G01	linear move at feed rate	**M00**	pause for the operator	**S / F / T**	speed / feed / tool no.

UNIT 3

Working with G-code

Small edits at the edges of a program are routine — editing the geometry in the middle is CAM's job.

A short program, annotated

```
(SLOT — 6MM 2-FLUTE ENDMILL)
G21 G90 G17     mm, absolute, XY plane
G54             work offset = part zero
G0 Z5.0         rapid to safe height
M3 S10600       spindle on CW, 10600 rpm
G0 X0 Y0        rapid above the start
G1 Z-2.0 F300   feed down to depth
G1 X50.0 F640   cut the slot at feed
G0 Z5.0         retract
M5              spindle off
M30             end of program
```

Safe to edit by hand

- Feed (F) and spindle speed (S) — ease off for chatter or burning
- Safe / clearance Z heights — after checking nothing is taller
- Work offset (G54→G55); add a pause (M00) or dwell (G04)
- Trim header/footer lines your machine doesn't use; drop an unused tool section

Regenerate from CAM — don't hand-edit

- Individual X / Y / Z cutting values, or arc I / J centres
- Switching units (G20 ↔ G21) partway through
- Cutter compensation (G41 / G42) offsets

Prove it before it cuts: re-simulate the whole program, then run it once as an air pass with a hand on the feed-hold — a G-code error is found by the machine at full speed.

UNIT 3

Laser Cutting

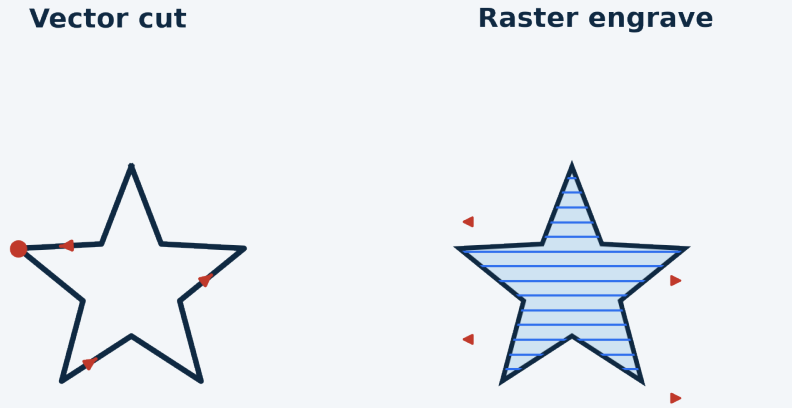
Cuts flat sheet by burning or vaporizing a narrow line through it — fast and precise in 2D, with its own rules.

Kerf — the cut removes a finite width of material — account for it on tight-fitting parts

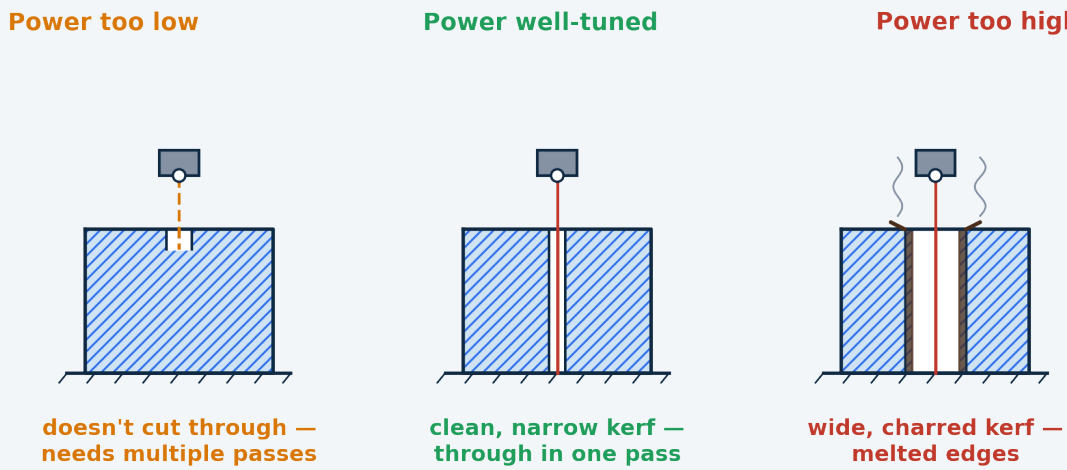
Material limits — wood, acrylic, paper, and cloth cut well

Small-feature limits — minimum hole size and spacing relative to material thickness

Vector Cut vs. Raster Engrave



How Laser Power Shows Up in the Cut



Never cut: PVC and other chlorinated plastics must never be cut on a laser — they release chlorine gas.

UNIT 3

DFM Design Considerations

Shape the part so the chosen process can make it well, at reasonable cost, without special effort.

1 Match design to process

uniform walls for printing/casting, generous radii for machining, flat unfolded geometry for laser

2 Minimize setups

features on one face are cheaper than features needing re-fixturing

3 Tolerance only what matters

tight tolerances drive cost — apply to mating/functional features

4 Use standard stock & tooling

standard sizes and drill/tool radii cost less, arrive faster

5 Design for prototype AND product

what's fine for a one-off print may not survive production

UNIT 3

Unit 3 Wrap-Up

1 Manufacturing-ready files

correctly exported for the chosen process

2 GD&T documentation

for at least one process, showing what you'd control and why

3 A DFM-reviewed design

ready to hand off to a shop process

You'll produce: correctly stated tolerances and fits, manufacturing-ready files and GD&T documentation demonstrating your design-for-manufacturing reasoning, plus reading and safely editing a G-code file.
Assessed by: Unit 3 Competency Check (Week 16). The physical prototype build & presentation is graded separately as the course Final Exam.

QUICK REFERENCE

All Units at a Glance

Keep this page open while you work — every unit's timing, focus, and assessment in one place.

Foundations	Before Unit 1	Define the problem, design intent, top-down vs. bottom-up	—
Unit 1 — CAD Fundamentals	Weeks 4–8	Controlled sketches, multiview drawing, first parametric model	Competency Check, Wk 8
Unit 2 — Design for Assembly	Weeks 8–11	Threads & fasteners, mates, joining methods, DFA principles	Competency Check, Wk 11
Unit 3 — Design for Manufacturing	Weeks 12–16	Tolerancing & fits, stackups, GD&T, manufacturing files, G-code, additive/subtractive/laser, DFM, safety	Competency Check, Wk 16
Final Exam — Prototype Project	Week 17	Team-built physical prototype, manufacturing docs, presentation	30% (w/ prototype)